Dataset Composition and Sample Selection for Reliable Fine Tuning

Findings and Scientific Interpretation

1 Experimental Setup

The following is a summary of the experimental design to allow the results in Section 2 to be evaluated without the use of the notebook. All experiments fine-tuned one pretrained Vision Transformer (ViT), ViT B/16 (the base model size with image patches of 16 by 16 pixels), which was pretrained on ImageNet 21k and then fine-tuned on ImageNet 1k. CIFAR100 served as the in distribution (ID) dataset. For validation, 50 images per class were used, while the remaining 45,000 images were used as a selection pool. The out of distribution (OOD) sets were CIFAR10 as near OOD, which means it contains natural images that resemble the ID classes, together with SVHN, Textures, MNIST and Gaussian noise as far OOD sets. Training subsets contained 5, 10, 25, 50, 75 and 100% of the pool and were chosen either at random or by representative sampling. Representative sampling was performed using k means clustering on the pretrained embeddings and the image closest to the cluster centre was retained. A second, long tailed pool of 9,754 images, in which the largest class has 112 times more images than the smallest, supported the comparison between naturally imbalanced and class balanced subsets.

Every condition used three seeds and three training protocols: 1,000 optimisation steps, a fixed budget of 3 epochs and a fixed budget of 10 epochs. Further runs were conducted with 250 and 3,000 steps, with fixed subset sizes. Four scores were used for OOD detection. The maximum softmax probability (MSP) and the Energy score are based on the output logits, whereas the Mahalanobis score and the k-nearest neighbour (kNN) score are based on the distance of an image from the training images in feature space. The evaluation reported ID accuracy, worst class accuracy, area under the receiver operating characteristic curve (AUROC), false positive rate (FPR) at 95% true positive rate (TPR), expected calibration error (ECE) and feature drift. AUROC is the probability that a random ID image receives a higher score than a random OOD image, and feature drift is one minus the linear centred kernel alignment (CKA) similarity between the fine-tuned and pretrained features. The study ran a total of 633 fine tuning runs. Their results are interpreted in the next section.

2 Findings and Interpretation

This section reports the main findings, assesses the strength of the evidence and identifies the resulting limitations and hypotheses.

2.1 Most Important Findings

2.1.1 Effect of Class Balance

Class balance had the largest effect of the three dimensions studied. At 5% of the long tailed pool with random selection, a class balanced subset raised ID accuracy from 45.2% to 76.3% and few shot class accuracy by 65.8 percentage points. In addition, it raised the Energy AUROC on CIFAR10 by 0.11. However, the new controls show that most of this benefit does not require selecting balanced data. When the complete long tailed pool was used with a class weighted loss or with balanced sampling, ID accuracy rose from 77.2% to about 85%. This equals the best balanced subset, which reached 84.6% at 25%. At the same time, the process controls left the Mahalanobis AUROC almost unchanged, with differences below 0.003, while the Energy AUROC rose by up to 0.06. This indicates that balancing primarily corrects the output of the classifier and not the learned features. The selection effect is

caused by a different factor at small sizes. A natural 5% subset contained only about 72 of the 100 classes, so reweighting cannot help classes that are absent, and the weighted loss recovered almost none of the gain. Therefore, the benefit of balanced selection at small sizes is primarily a class coverage benefit.

2.1.2 Relation Between ID Accuracy and OOD Detection

The experiments confirm that ID accuracy and OOD detection improve together for semantically similar OOD data. Across all protocols, the within size Spearman correlation between ID accuracy and AUROC on CIFAR10 was 0.94 for Energy and 0.92 for Mahalanobis, and similar values held for Textures. The MSP score led to the same conclusion, with a within size correlation of 0.97 on CIFAR10, and it agreed with Energy on the direction of every class balancing effect on CIFAR10 and SVHN. This was true for all training lengths, fine tuning regimes and balancing methods. The exception is MNIST. On MNIST, the feature based scores correlated negatively with ID accuracy, with magnitudes between 0.64 and 0.70. Training the same subset for longer reduced the AUROC (both Mahalanobis and kNN) on MNIST in all 11 subset conditions. In addition, fine tuning reduced the Mahalanobis AUROC on MNIST below the value of the pretrained features, which was 0.89, to between 0.73 and 0.81. The same fine tuning, however, increased the Mahalanobis AUROC for CIFAR10 by approximately 0.10 to 0.13. This is supported by the fine tuning regimes. A frozen backbone and a reduced learning rate both maintained MNIST detection at approximately 0.90, but reduced the Mahalanobis AUROC for CIFAR10 to between 0.81 and 0.90. Thus, the trade-off appears between types of OOD data rather than between ID accuracy and OOD detection in general.

The mechanism diagnostics provide a possible explanation for this pattern. Within each subset size, a lower ratio of within class to between class feature variance correlated with a higher CIFAR10 AUROC, with magnitudes between 0.79 and 0.89. This means that fine tuning makes classes more compact, which helps to separate semantically similar OOD images. At the same time, it appears to discard low level information that distinguishes digit images from natural images.

2.1.3 Dataset Size and Sample Selection

The impact of the size of the dataset differed substantially across protocols. Moving from 5% to 100% of the data raised ID accuracy by 4.0 points under 1,000 steps, by 26.5 points under 10 epochs and by 77.5 points under 3 epochs. The step sweep accounts for this difference. At 5%, the original protocol had already converged on small subsets, as training for 3,000 steps yielded the same accuracy as training for 1,000 steps (86.5%). Consequently, the baseline size effect reflects the information in the data, while the fixed epoch protocols mainly measure undertraining. Representative sampling, however, did not appear to offer any substantial benefit on the balanced pool. Under all protocols, its accuracy was within 0.2 points of random selection. However, on the long tailed pool at 5%, it raised accuracy from 45.2% to 53.5% because it covered about 80 classes instead of 72. Again, this suggests that class coverage is the active factor.

2.2 Strength of the Evidence

The conclusions differ in the strength of the experiments supporting them. There are four conclusions that are well supported. The first is that class balance improves ID accuracy, few shot accuracy and CIFAR10 detection, which was significant in all ten selection and size conditions. The second is that ID accuracy and near OOD detection improve together, and this held in more than 400 control runs and in all protocols. The third is that representative sampling and random sampling yield the same accuracy when sampling a balanced pool. The fourth is that training for 1,000 steps had converged at small subset sizes.

Other conclusions are tentative. The MNIST trade-off is based on a single far OOD dataset, and the Energy AUROC on MNIST was highly sensitive to seed. The mechanism results are correlative and not causal. Furthermore, the separation between selection and process effects is not complete at small sizes, as discussed in Section 2.1.1, because class coverage is confounded with class balance. Lastly,

the comparison to the ImageNet 21k backbone is not yet informative, as Section 2.3 explains.

2.3 Methodological Limitations

Several limits apply to the interpretation of the results:

- Each condition used only three seeds, and the significance tests did not correct for the several hundred comparisons performed. Therefore, some individually significant differences may be false positives.
- The study used one ID dataset and one architecture with fixed hyperparameters. The training configuration shared by all runs did not suit the ImageNet 21k backbone, which ended training with a loss of 1.55 and an ECE of about 0.6.
- The 3 epoch protocol gave the smallest subsets only 53 steps, or 11 steps on the long tailed pool, so those models barely learned. No protocol trained each condition until the validation accuracy stopped improving.
- The kNN and training based Mahalanobis detectors were fitted on up to 100 images per class. The number of images in their fitting set thus increased from 2,250 images at 5% to 10,000 images at 25% and above. Part of their size effect may reflect this difference rather than the model.
- The long tailed pool was artificially created from the balanced CIFAR100 data with one seed, and the MNIST and noise images were upscaled from 32 to 224 pixels.

These limitations influence the hypotheses proposed in Section 2.4.

2.4 Hypotheses and Proposed Experiments

The results suggest three pairs of competing hypotheses, each of which can be tested in a targeted experiment.

The first pair concerns the MNIST trade-off. Hypothesis A states that fine tuning replaces low level visual information with semantic information, which harms the detection of far OOD data with different low level statistics. Hypothesis B is that the effect is only present for MNIST because the images are grayscale and highly enlarged. Measuring Mahalanobis AUROC at intermediate transformer layers and on additional far OOD sets, including grayscale versions of natural images, would separate the two. If the loss is only in the last few layers and for all low level shifts, then Hypothesis A is supported.

The second pair relates to the balancing effect. Hypothesis C is that the balanced selection helps primarily because it contains all classes, and Hypothesis D is that the proportions of the classes themselves are important. A comparison between balanced subsets and natural subsets that are forced to contain every class would distinguish them. If the forced coverage subsets close up most of the gap, then Hypothesis C is supported.

The third pair relates to where balancing acts. Hypothesis E says that balancing improves logit based detection only by correcting the classifier, whereas Hypothesis F says that it also changes the features. Retraining only the classifier with a balanced loss on features from a model trained on natural data would test this. If retraining the classifier alone gives the same Energy AUROC gain, Hypothesis E is supported, which is consistent with the unchanged Mahalanobis results in Section 2.1.1. These experiments would explain why the observed effects occur, and not just where they occur.

3 Response to the Supervisory Feedback

This section lists how the implementation addressed the first three points of the feedback.

- Training protocol: the study added fixed budgets of 3 and 10 epochs for every subset size, a

step sweep at fixed subset size and a regression that separates the effect of more images from the effect of more steps.

- OOD evaluation: the study analysed the existing MSP results against Energy, added Mahalanobis detectors fitted on the training subset and on the validation set, added a kNN detector and added Textures, MNIST and Gaussian noise as OOD sets. A separate analysis was conducted to test whether the absence of a trade-off holds across scores, protocols and fine tuning regimes.
- Class imbalance: the study added a class weighted loss and balanced sampling, applied both to the original subsets and to the complete long tailed dataset, and tested four further balancing methods. The effect of selecting balanced data was separated from the effect of balancing the training process by paired contrasts.

These additions provide the evidence for the findings reported in Section 2.